VIGNESH SWAMINATHAN

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PROFESSIONAL SUMMARY

AI/software engineer with 2+ years of hands-on experience in backend development, cloud application engineering, and large-scale AI system evaluation across Python and JavaScript environments. Master's degree in computer science with practical experience in large language model integration (Claude, OpenAI GPT, Gemini), cloud platforms (Azure, AWS), and machine learning using tools such as TensorFlow and PyTorch. Skilled in RESTful API design, CI/CD pipelines, distributed data processing, and full-stack frameworks, with hands-on experience delivering production-grade software. Proven track record of accelerating engineering workflows, maintaining system reliability and documentation, strong attention to detail, clear communication, and a mindset focused on continuously learning emerging AI tools and technologies.

EDUCATION

Boston University | Boston, MA
Master of Science in Computer Science | GPA: 3.65/4.0 ***Jan 2026***
Relevant Coursework: Software Engineering, Cybersecurity, Deep Reinforcement Learning, Big Data Analytics, Data Science

Visvesvaraya Technological University | Bengaluru, IN
B.E. in Information Science | GPA: 3.60/4.0 ***May 2024***
Relevant Coursework: Data Structures, AIML, Software Testing, Computer Networks and Security, Storage Area Networks

TECHNICAL SKILLS

AI/ML & LLM Engineering: Python, LLMs, Retrieval-Augmented Generation (RAG), LangChain, OpenAI GPT (Prompt Engineering, API Integration), Claude, Gemini, TensorFlow, PyTorch, Scikit-learn, NLP, Deep Learning, Model Fine-Tuning, Hugging Face Transformers, Vector Databases, AI Agents
Data Engineering & Analytics: Pandas, NumPy, ETL/ELT Pipelines, Data Modeling, Data Warehousing, Hadoop, Apache Spark, PySpark, Databricks, Power BI, Data Analysis, Data Visualization, Large-Scale Data Processing, Apache Kafka, Apache Airflow
Software & Full-Stack Development: Python, Java, JavaScript, C, C++, HTML, CSS, PHP, ASP.NET, React.js, Django, Node.js, RESTful APIs, GraphQL, API Integration, Microservices Architecture, Object-Oriented Design
Cloud, DevOps & Security: Azure, AWS (EC2), Docker, Jenkins, Kubernetes, Terraform, CI/CD Pipelines, GitHub, Git, Jira, ServiceNow, Version Control, Linux Systems, Identity & Access Management, Endpoint Security, Vulnerability Management
Databases & Tools: MySQL, PostgreSQL, MongoDB, Redis, NoSQL, Office 365, Android Studio, Flutter, Figma, Metasploit, OpenSSL, Agile/Scrum Methodologies

PROFESSIONAL EXPERIENCE

Boston University | Boston, MA
Research Fellow: Law, AI Governance & Tech Policy ***July 2026 - Present***
- Conducted large-scale empirical evaluation of AI systems by benchmarking hallucination rates, citation accuracy, and reasoning reliability using **Python, REST APIs, and Git**, and structured annotation frameworks, improving evaluation consistency by **30%** overall across teams
- Designed and validated **AI verification workflows** leveraging LLM APIs, Court Listener API, and data analysis techniques to enhance citation grounding, model reliability, and trustworthy AI adoption across legal technology research and academic publications and top journals

Cloud Application Developer ***Feb 2026 - May 2026***
- Engineered **full-stack cloud-based solutions** for onboarding workflows and automated data pipelines to ingest and transform legacy SharePoint data into Dataverse, improving efficiency by **40%** and strengthening centralized Azure data management systems and pipelines
- Designed automated **CI/CD pipelines** for Power Fx solutions while implementing security best practices, including role-based access control and data loss prevention policies, to ensure secure and scalable application delivery across the development lifecycle and compliance reviews

Information Technology Analyst ***Sep 2024 - Jan 2026***
- Led development and automation initiatives on the **ServiceNow platform**, designing AI-assisted backend workflows and automation scripts to streamline ticket triage, reporting, and resolution for enterprise-scale systems supporting daily operations across departments and teams
- Maintained cloud-native backend tools and dashboards using **Microsoft Azure, Power BI, and Office 365** integrations, optimized MySQL databases and API documentation, reducing average ticket resolution time by **25%** and improving system reliability and uptime metrics

Jio Platforms Limited (Reliance Industries) | Bengaluru, IN
Android Developer Intern (Project: AVNI OS) ***Aug 2023 - Dec 2023***
- Collaborated with engineers and design teams to translate product requirements into **Bluetooth-enabled**, scalable mobile features, ensuring alignment with backend architecture and API integrations, improving feature delivery accuracy by **20%** across product releases and sprints
- Performed automated testing and triaged bugs with engineers, managed project development via **Microsoft Azure, Android Studio, and Flutter**, and reviewed UI/UX design based on Figma prototypes, increasing workflow efficiency by **15%** across development sprints overall

PROJECTS

- **AI-Text Detection Benchmarking:** Built and de-identified a corpus of human- and AI-generated legal exam answers to benchmark AI-text detector accuracy and evaluate adversarial evasion techniques against RLHF-based detection fingerprints
- **Menu Match:** A web app for restaurant discovery and reviews based on dietary preferences, integrating Google APIs, NLP, and ML recommendations, deployed with Docker, AWS EC2, Jenkins, GitHub for CI/CD, and managed via JIRA
- **Collider Scope:** Built distributed data processing pipelines using PySpark and Databricks to analyze large-scale particle physics datasets and develop scalable machine learning workflows, achieving an AUC score above 0.85
- **Safe Drive - Traffic Accident Severity Prediction:** Processed and analyzed 7.7 million accident records using Apache Spark, PySpark, and AWS. Built scalable ETL pipelines for data ingestion, cleansing, and model training
- **Law-Run Kernel:** Developed and maintained a custom Linux 4.9 based kernel for Qualcomm Snapdragon 845 devices, combining CAF sources with upstream Linux patches and a security driver, published on GitHub for community usage

CONFERENCE & PUBLICATIONS

- **AI Based Security Surveillance System – An Overview:** Accepted and published by ieeeForum in association with the Institute of Research and Journals, International Conference on Advances in Artificial Intelligence and Computer Science (ICAAICS-2024), Chennai, IN
- **AI and Blockchain in Food and Beverage – Marketing 5.0 Approach:** Published in the Springer LNNS series, 7th International Conference on Innovative Computing and Communication 2024, organized by University of Delhi, NIT Patna & University of Valladolid, Spain

CERTIFICATIONS

- Introduction to Programming with Python and Java – University of Pennsylvania
- Google Data Analytics Specialization